Serverless

요금

	Flex	Active
비유	택시 (필요할 때만)	전속 기사 (항상 대기)
장점	안 쓸 때 돈 안 냄	빠른 응답, 할인 요금
단점	시작할 때 약간 느림	안 써도 돈 나감

Flex Workers

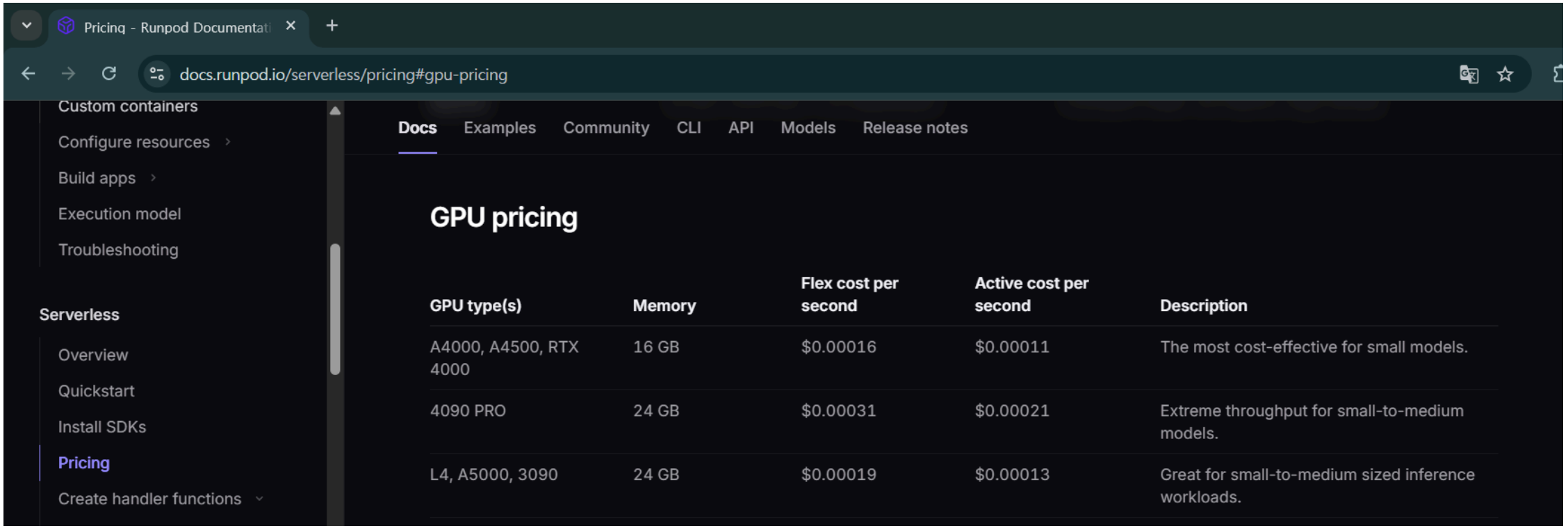
- **일 안할 땐 꺼져 있음** (Scale to zero)
- 요청이 오면 켜지고, 끝나면 다시 꺼짐
- **가격이 더 비쌈** (기본 요금)
- 트래픽이 들쭉날쭉할 때 유리

Active Workers

- **24시간 계속 켜져 있음**
- 요청이 오면 바로 처리 (대기 시간 거의 없음)
- **20~30% 더 저렴** (항상 쓰니까 할인)
- 트래픽이 꾸준히 많을 때 유리

GPU pricing

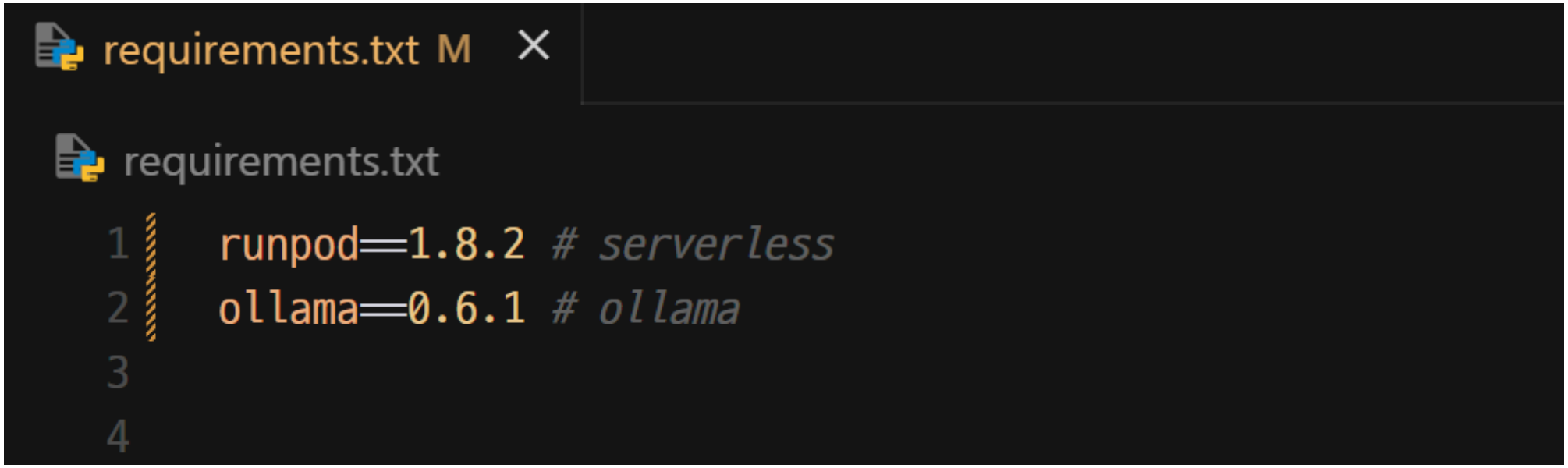
> Runpod 정책에 따라 매번 금액이 다를 수 있으므로, 공식 홈페이지에서 확인해야 함.



예제: Flex Workers with Ollama

1. 코드 확인

라이브러리 정의


requirements.txt M
requirements.txt
runpod==1.8.2 # serverless
ollama==0.6.1 # ollama

runpod serverless 코드

handler.py M

handler.py > ...

```
29
30  def ensure_ollama_server() → None: ...
45
46
47  def ollama_chat(prompt: str, model: str, system: Optional[str] = None) → str: ...
57
58
59  def handler(event): ...
70
71
72  if __name__ == "__main__":
73      ensure_ollama_server() # ollama 서버 준비
74
75      runpod.serverless.start({"handler": handler}) # serverless 시작
76
```

올라마 모델 다운로드

download_ollama_models.sh

```
14    while ! ollama list >/dev/null 2>&1; do
20        sleep 1
21    done
22
23    echo "[build] Pulling model: $MODEL"
24    ollama pull "$MODEL"
25
26    echo "[build] Stopping ollama serve..."
27    kill "$PID" >/dev/null 2>&1 || true
28
29    echo "[build] Model download completed."
```

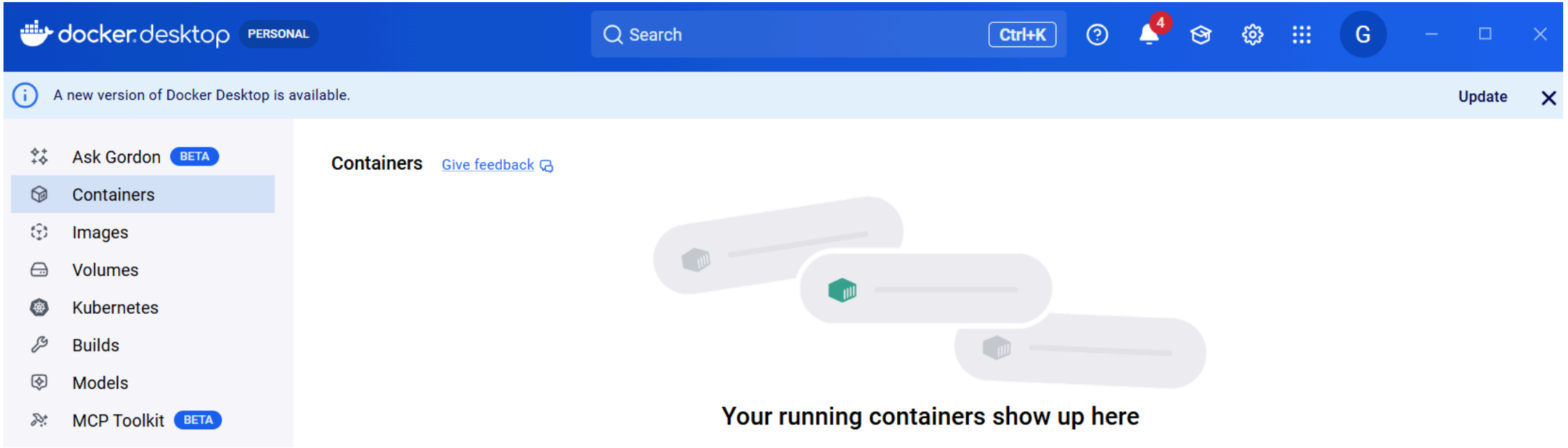
도커파일

Dockerfile

```
18  # 빌드 중 임시로 서버를 띄운 뒤 모델을 받습니다.
19  COPY download_ollama_models.sh /download_ollama_models.sh
20  RUN sed -i 's/\r$//' /download_ollama_models.sh  && chmod +x /download_ollama_models.sh
21
22  # Copy your handler file
23  COPY handler.py /
24  COPY test_input.json /
25  # Install dependencies
26  COPY requirements.txt /requirements.txt
27  RUN python3 -m pip install --no-cache-dir --break-system-packages \
28      -r /requirements.txt
29
30  # Start the container
31  # Base image has ENTRYPOINT ["ollama"], so override it.
32  ENTRYPOINT ["python3", "-u", "/handler.py"]
```

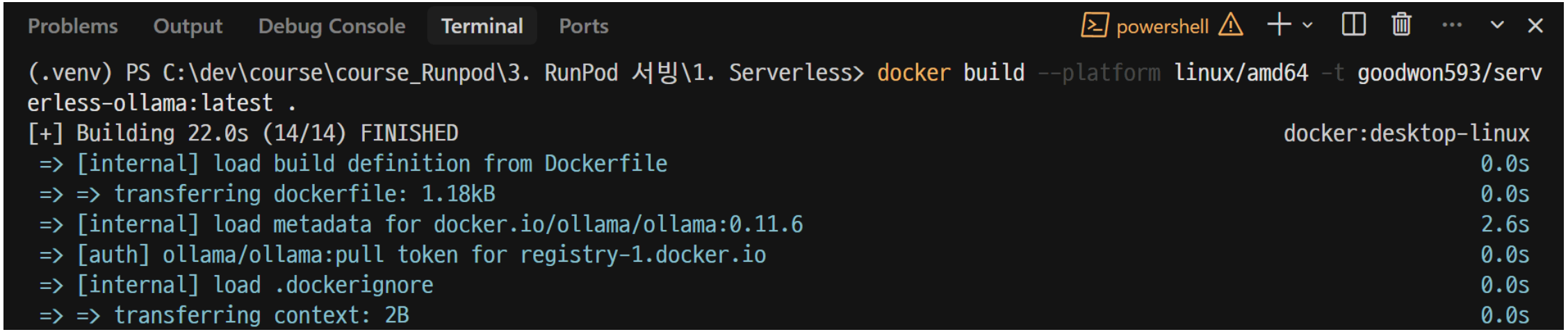
2. Docker

단계1: Docker Server 실행 및 로그인

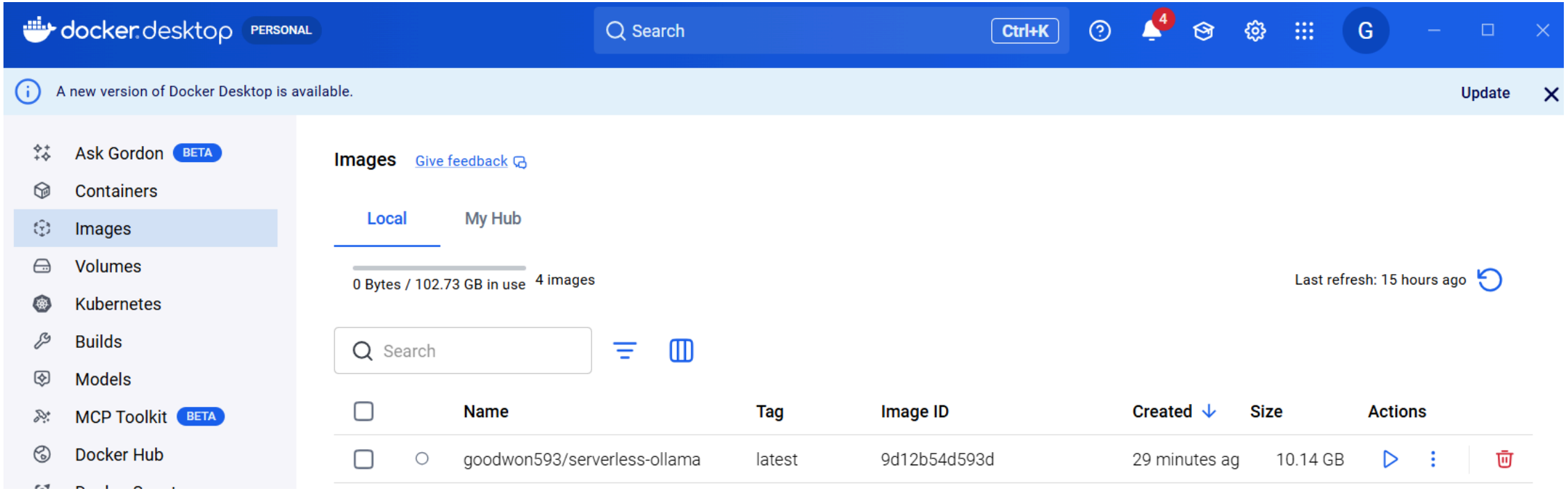


단계2: Docker Image 생성

```
# 도커파일이 있는 폴더에서 실행
docker build --platform linux/amd64 -t [YOUR_USERNAME]/serverless-ollama:latest .
```



생성된 이미지 확인


docker desktop
PERSONAL
Search
Ctrl+K
4
G
A new version of Docker Desktop is available.
Update
Ask Gordon
BETA
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit
BETA
Docker Hub
Images
Give feedback
Local
My Hub
0 Bytes / 102.73 GB in use
4 images
Last refresh: 15 hours ago
Search
Name
Tag
Image ID
Created
Size
Actions
goodwon593/serverless-ollama
latest
9d12b54d593d
29 minutes ag
10.14 GB

단계3: 생성된 Docker Image 테스트

> test_input.json을 통해 테스트 진행됨.

test_input.json

```
{
    "input": {
        "model": "gemma3:4b",
        "system": "You are a helpful assistant.",
        "prompt": "Hey there! 한글로 짧게 인사해줘."
    }
}
```

테스트

```
docker run -it [YOUR_USERNAME]/serverless-ollama:latest
```

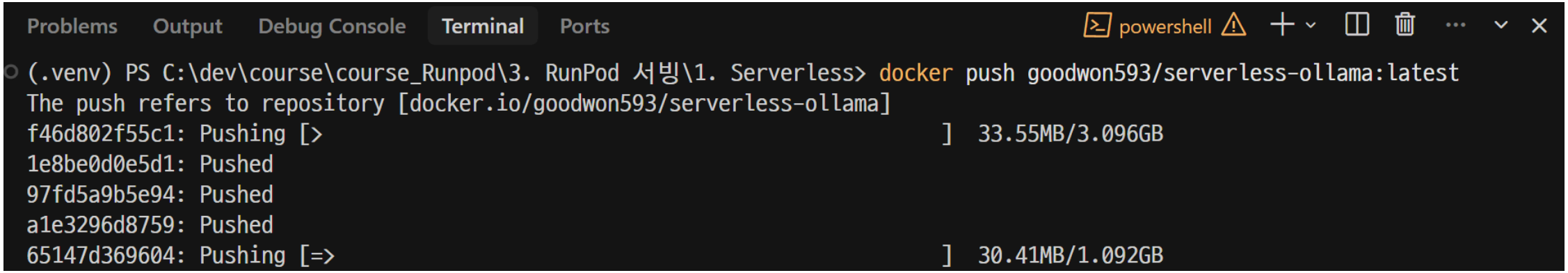
```
(.venv) PS C:\dev\course\course_Runpod\3. RunPod 서빙\1. Serverless> docker run -it goodwon593/serverless-ollama:latest
Ollama 서버가 꺼져 있어 `ollama serve`를 시작합니다.
_client.py          :1025 2026-04-01 01:02:45,167 HTTP Request: GET http://localhost:11434/api/tags "HTTP/1.1 200 OK"
--- Starting Serverless Worker |  Version 1.8.2 ---
INFO   | Using test_input.json as job input.
DEBUG  | Retrieved local job: {'input': {'model': 'gemma3:4b', 'system': 'You are a helpful assistant.', 'prompt': 'Hey the
re! 한글로 짧게 인사해줘.'}, 'id': 'local_test'}
INFO   | local_test | Started.
_client.py          :1025 2026-04-01 01:02:50,098 HTTP Request: POST http://localhost:11434/api/chat "HTTP/1.1 200 OK"
DEBUG  | local_test | Handler output: 안녕하세요! 😊

(Annyeonghaseyo!) - Hello!

How can I help you today?
DEBUG  | local_test | run_job return: {'output': '안녕하세요! 😊 \n\n(Annyeonghaseyo!) - Hello!\n\nHow can I help you today
?'}
INFO   | Job local_test completed successfully.
INFO   | Job result: {'output': '안녕하세요! 😊 \n\n(Annyeonghaseyo!) - Hello!\n\nHow can I help you today?'}
INFO   | Local testing complete, exiting.
(.venv) PS C:\dev\course\course_Runpod\3. RunPod 서빙\1. Serverless>
```

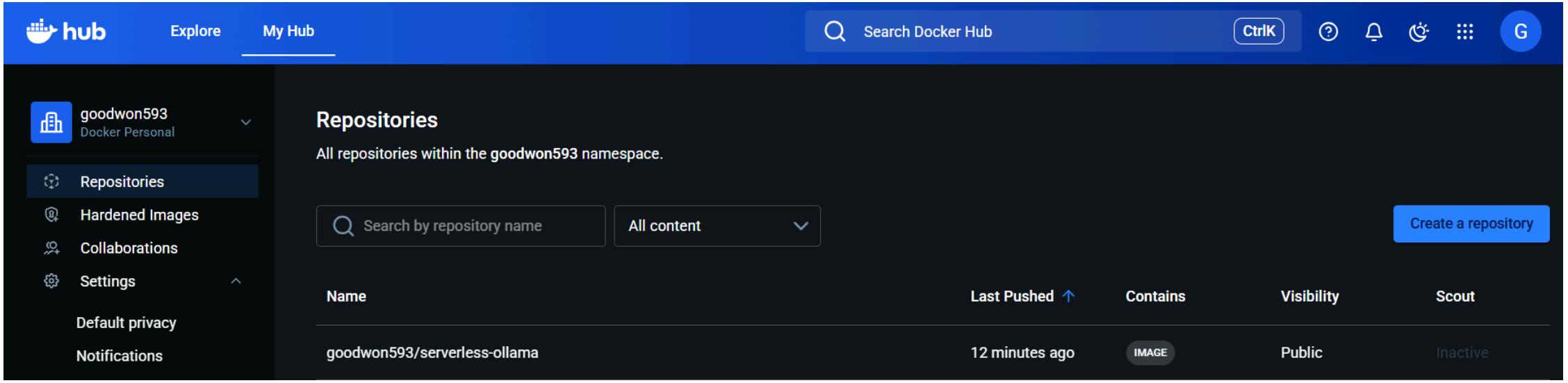
단계4: Docker Hub 배포

```
docker push [YOUR_USERNAME]/serverless-ollama:latest
```



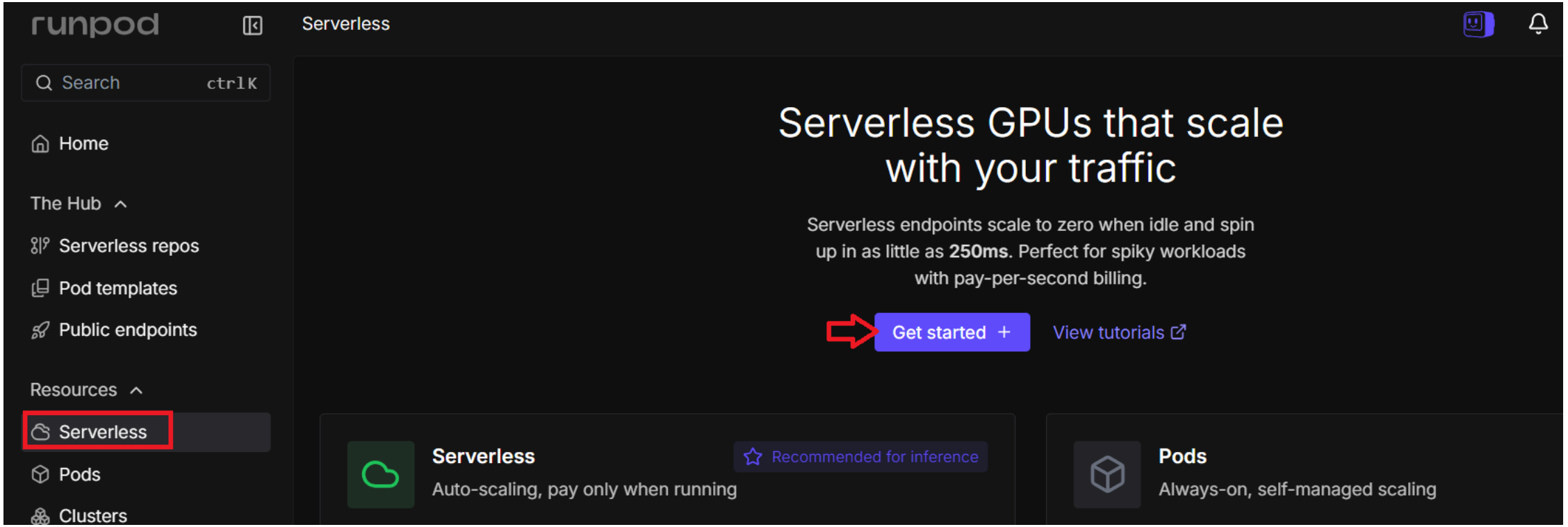
단계5: Docker Hub

> 결과 확인

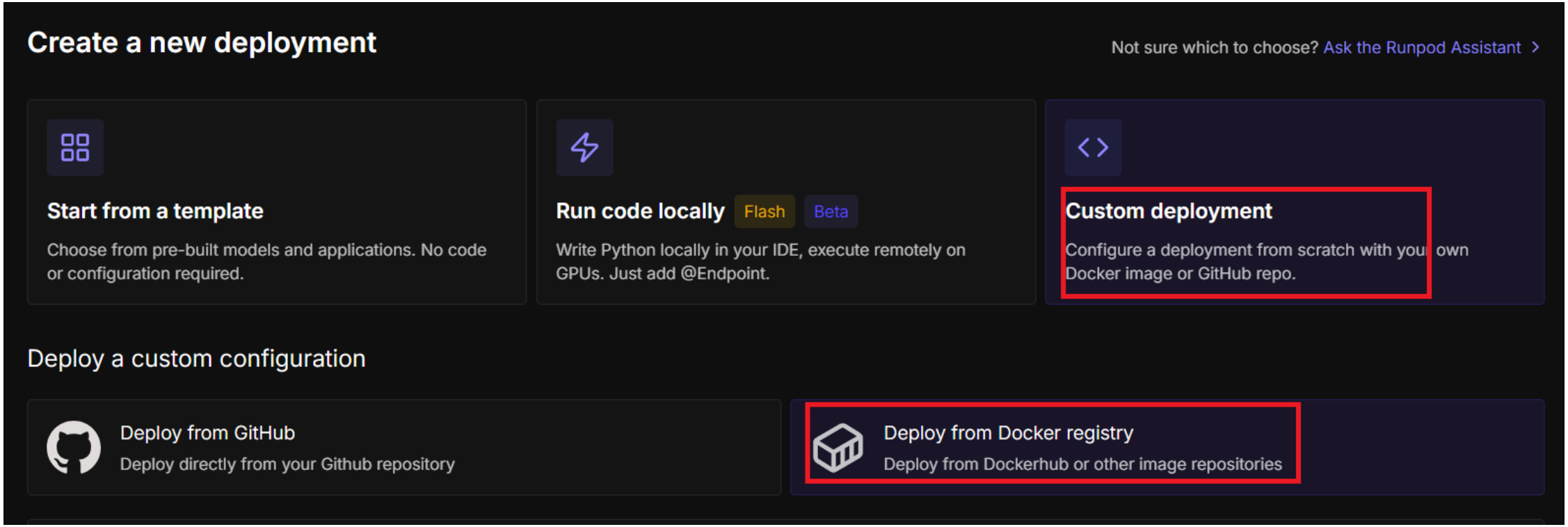


3. Runpod's Serverless

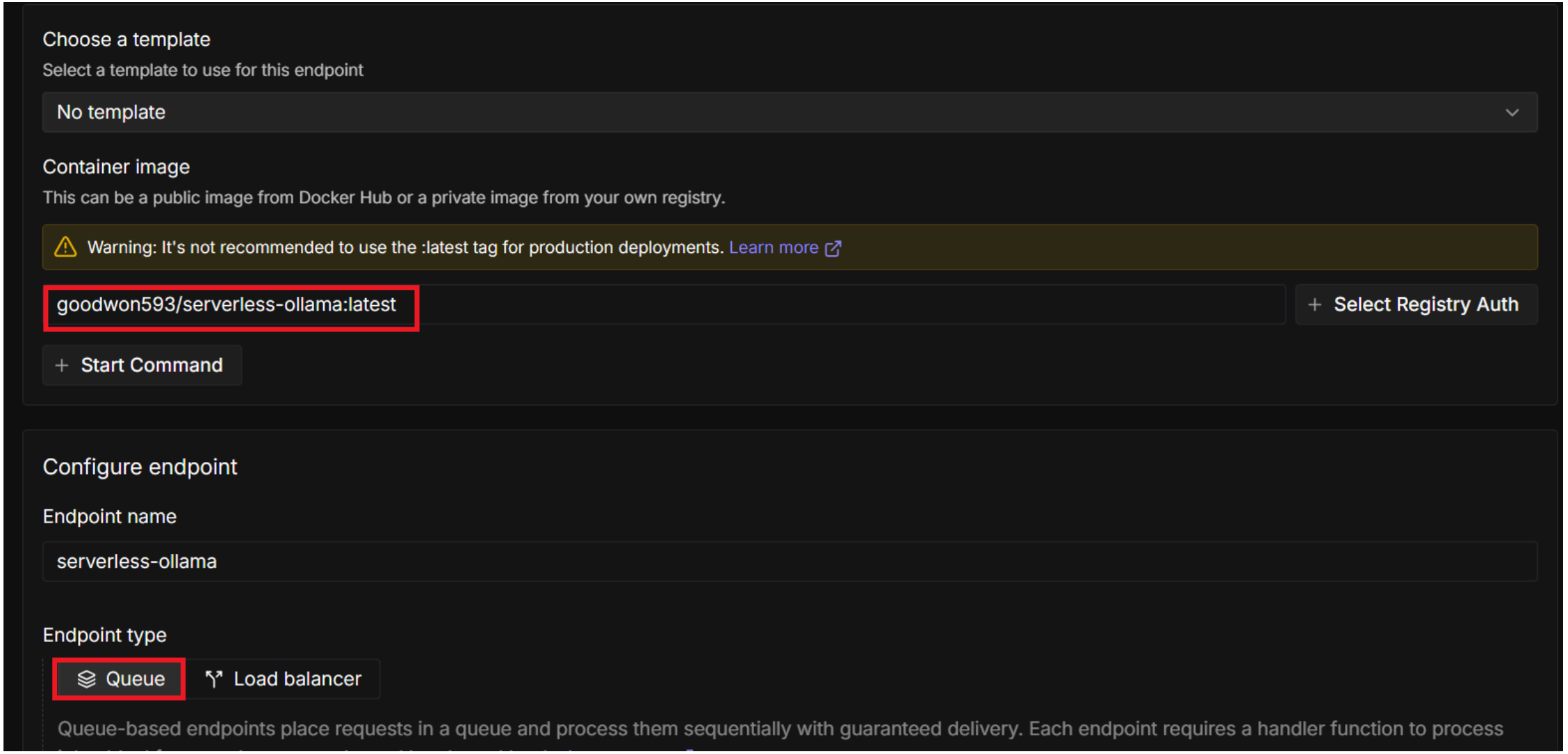
단계1: Serverless > Get started



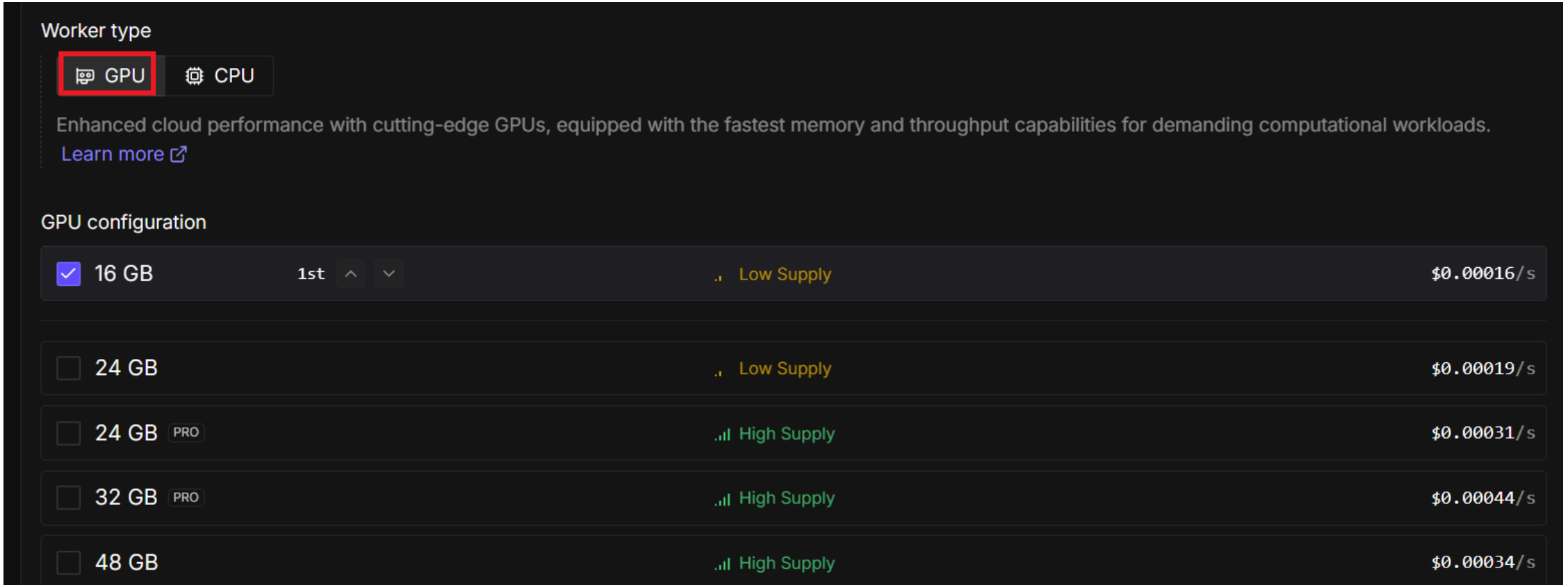
단계2: Create a new deployment



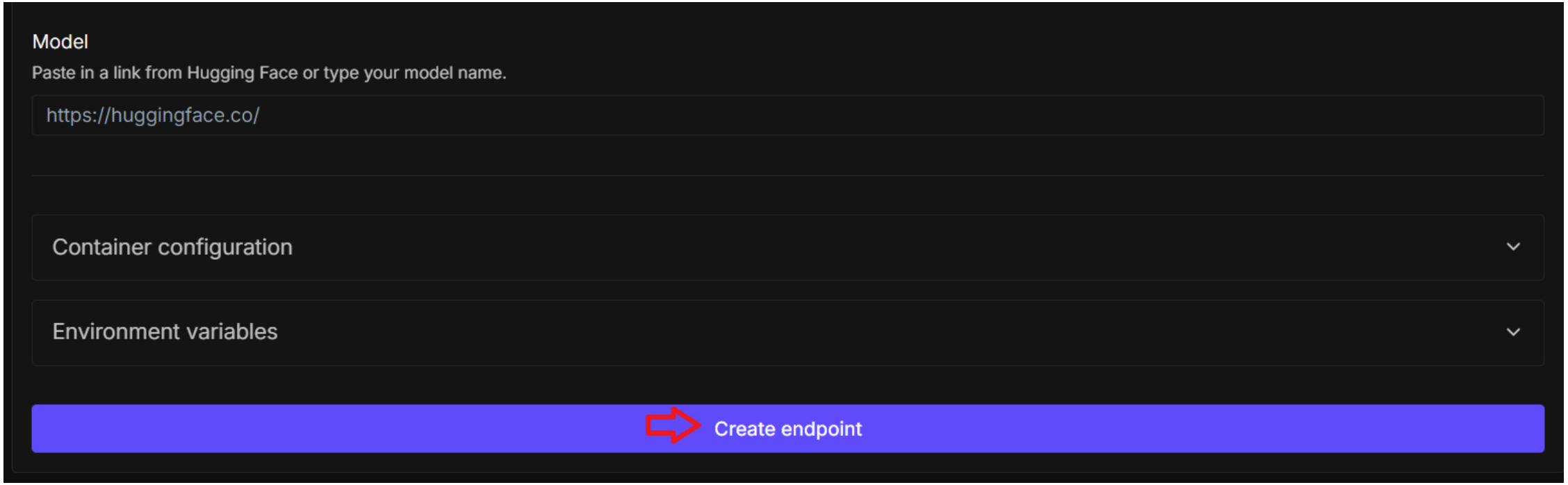
Container image



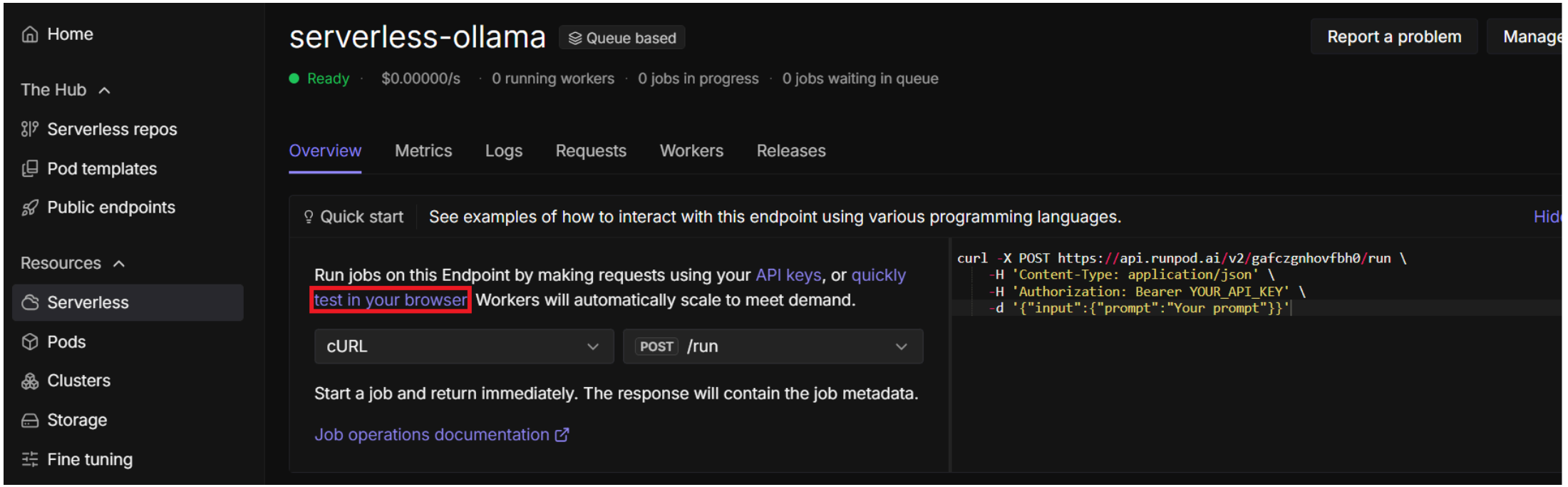
Worker type


Worker type
GPU
CPU
Enhanced cloud performance with cutting-edge GPUs, equipped with the fastest memory and throughput capabilities for demanding computational workloads.
Learn more
GPU configuration
16 GB
1st
Low Supply
$0.00016/s
24 GB
Low Supply
$0.00019/s
24 GB PRO
High Supply
$0.00031/s
32 GB PRO
High Supply
$0.00044/s
48 GB
High Supply
$0.00034/s

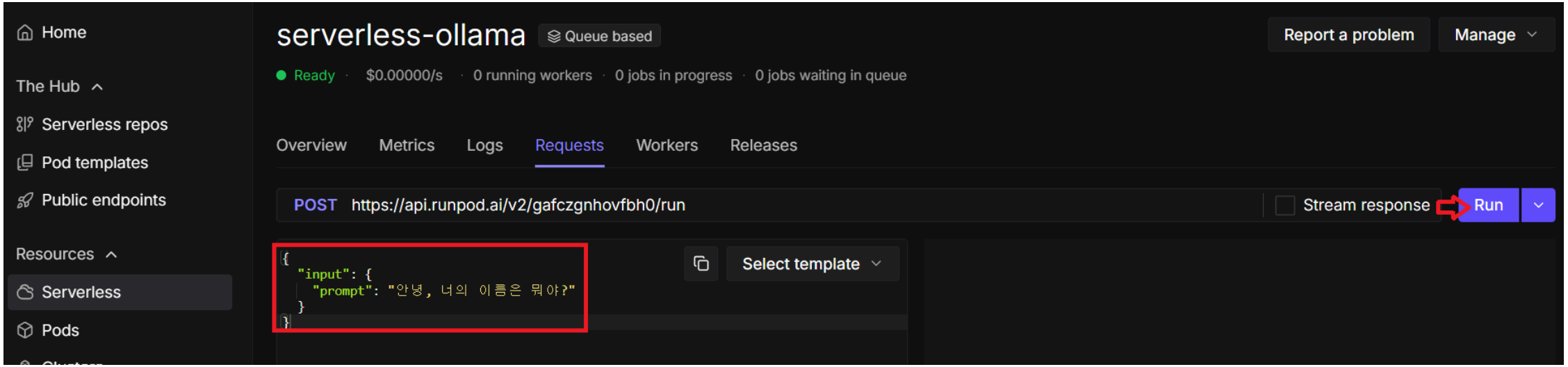
Create endpoint



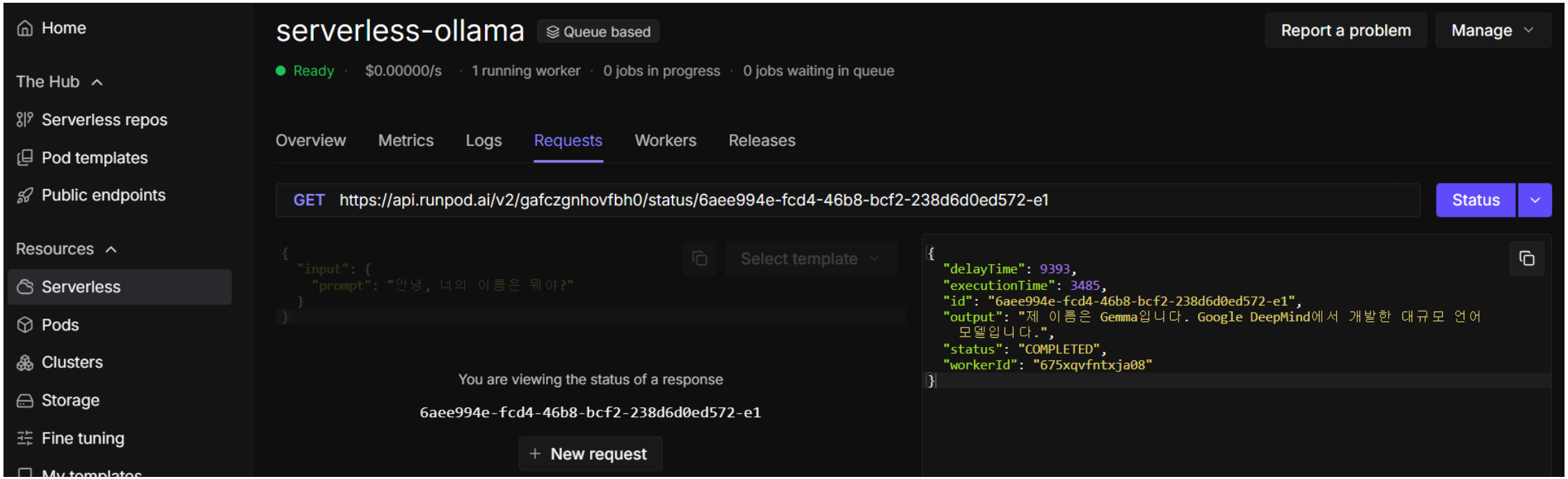
단계3: Serverless 테스트



테스트 실행


Home
The Hub
Serverless repos
Pod templates
Public endpoints
Resources
Serverless
Pods
serverless-ollama
Queue based
Report a problem
Manage
Ready
$0.00000/s
0 running workers
0 jobs in progress
0 jobs waiting in queue
Overview
Metrics
Logs
Requests
Workers
Releases
POST
https://api.runpod.ai/v2/gafczgnhovfbh0/run
Stream response
Run
Select template
{
"input": {
"prompt": "안녕, 너의 이름은 뭐야?"
}
}

테스트 결과


Home
The Hub
Serverless repos
Pod templates
Public endpoints
Resources
Serverless
Pods
Clusters
Storage
Fine tuning
serverless-ollama
Queue based
Report a problem
Manage
Ready · $0.00000/s · 1 running worker · 0 jobs in progress · 0 jobs waiting in queue
Overview
Metrics
Logs
Requests
Workers
Releases
GET https://api.runpod.ai/v2/gafczgnhovfbh0/status/6aee994e-fcd4-46b8-bcf2-238d6d0ed572-e1
Status
{
"input": {
"prompt": "안녕, 너의 이름은 뭐야?"
}
}
Select template
You are viewing the status of a response
6aee994e-fcd4-46b8-bcf2-238d6d0ed572-e1
New request
{
"delayTime": 9393,
"executionTime": 3485,
"id": "6aee994e-fcd4-46b8-bcf2-238d6d0ed572-e1",
"output": "제 이름은 Gemma입니다. Google DeepMind에서 개발한 대규모 언어 모델입니다.",
"status": "COMPLETED",
"workerId": "675xqvfntxja08"
}